

Decision Tree, Gini Index & Information Gain

Slide 1: Introduction

Decision Tree

- Supervised Machine Learning Algorithm
 - Used for Classification and Regression
 - Works like a flowchart
 - Easy to understand and visualize
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Slide 2: Structure of Decision Tree

Components

- Root Node → Starting point
 - Decision Node → Condition checking
 - Branch → Result of condition
 - Leaf Node → Final prediction/output
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Slide 3: Real-Time Example

Should We Play Cricket?

Conditions:

- Weather
- Humidity
- Wind

Example Rules:

- Sunny + High Humidity → No Play
 - Overcast → Play
 - Rainy + Weak Wind → Play
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Slide 4: How Decision Tree Works

1. Take complete dataset
2. Find best feature for split
3. Split data into subsets
4. Repeat recursively
5. Stop when pure node is achieved

Goal: Reduce impurity and improve prediction accuracy.

Slide 5: Understanding Impurity

Pure Node

All data belongs to same class.

Impure Node

Mixed classes are present.

Decision Tree Uses:

- Gini Index
 - Entropy
 - Information Gain
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Slide 6: Gini Index

Definition

Gini Index measures impurity in dataset.

Important Points

- Lower Gini → Better split
 - Higher Gini → More impurity
 - Range: 0 to 1
 - Gini = 0 means perfect purity
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Slide 7: Gini Index Formula

Formula

$$Gini = 1 - \sum (p_i)^2$$

Where:

- p_i = Probability of each class

Example

70% Yes and 30% No

$$\begin{aligned} Gini &= 1 - (0.7^2 + 0.3^2) \\ &= 1 - (0.49 + 0.09) \\ &= 0.42 \end{aligned}$$

Slide 8: Entropy

Definition

Entropy measures randomness or uncertainty.

Important Points

- Higher entropy → More confusion
 - Lower entropy → More purity
 - Used in ID3 and C4.5 algorithms
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Slide 9: Entropy Formula

Formula

$$Entropy = - \sum p_i \log_2(p_i)$$

Why log base 2?

- Information is measured in bits
- Binary classification naturally uses base 2

- Standard in information theory
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Slide 10: Entropy Example

Example

50% Yes and 50% No

$$Entropy = -(0.5\log_2(0.5) + 0.5\log_2(0.5))$$

$$Entropy = 1$$

Meaning: Maximum uncertainty.

Slide 11: Information Gain

Definition

Information Gain measures reduction in uncertainty after splitting.

Formula

$$Information\ Gain = Parent\ Entropy - Weighted\ Child\ Entropy$$

Important Points

- Higher gain → Better feature
 - Decision tree selects highest information gain
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Slide 12: Information Gain Example

Example

Parent Entropy = 1

Weighted Child Entropy = 0.4

$$Information\ Gain = 1 - 0.4$$

$$= 0.6$$

Meaning: Weather is a good feature for splitting.

Slide 13: Gini Index vs Information Gain

Gini Index	Information Gain
Faster computation	Uses entropy
Used in CART	Used in ID3/C4.5
Measures impurity	Measures uncertainty reduction
Less computationally expensive	Slightly slower

Slide 14: Advantages of Decision Tree

- Easy to understand
 - Easy visualization
 - Handles numerical & categorical data
 - Less preprocessing required
 - Useful for feature selection
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Slide 15: Disadvantages of Decision Tree

- Can overfit
 - Sensitive to small changes
 - Large trees become complex
 - Sometimes less accurate than ensemble models
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Slide 16: Applications of Decision Tree

- Medical diagnosis
 - Loan approval systems
 - Fraud detection
 - Customer churn prediction
 - Student performance prediction
 - Recommendation systems
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Slide 17: Conclusion

- Decision Tree is simple and powerful
 - Gini Index measures impurity
 - Entropy measures uncertainty
 - Information Gain selects best feature
 - Widely used in real-world machine learning
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Slide 18: Thank You

Questions & Discussion